VIII Tutorial

Volatility clustering, ARCH and GARCH models

Exercise 1. Modelling EUR/USD volatility

Notebook: Volatility_Clustering.ipynb

Analyse daily EUR/USD log changes for 2021–2025 (1,249 observations), using official Federal Reserve quotations. Examine volatility clustering and ARCH(1) moments, then compare ARCH/GARCH models using AIC and BIC. Interpret residual diagnostics, compare fitted volatility with rolling and EWMA measures, and examine one- and two-step forecasts.

Relevant concepts

- Log changes; squared changes as a variance proxy; ACFs; volatility clustering.
- Conditional versus unconditional moments; Gaussian ARCH(1) variance and kurtosis; second- and fourth-moment conditions.
- ARCH LM test; ARCH/GARCH estimation; parameter restrictions; AIC and BIC.
- Variance persistence; standardised-residual diagnostics; limits of inference.
- Rolling and EWMA volatility; information timing; historical filters versus forecast evaluation.
- Parameter boundaries; bootstrap comparison; variance versus volatility forecasts; units and horizons.

To do. Choose an asset and repeat the analysis using daily prices from an official or primary source. Document the source, sample and price adjustments. Compare the ACFs of log returns and squared demeaned returns. Fit **ARCH(1) and GARCH(1,1)** with the same mean specification and sample; compare AIC and BIC. For your preferred model, plot conditional volatility, inspect the ACF of squared standardised residuals, and discuss persistence and remaining clustering. Compare the findings with EUR/USD.